

TED (10) 5083

Rev (2010)

Fifth Semester Diploma Examination in Polymer Technology 2012

Model Question Paper

CHEMICAL ENGINEERING

I

PART-A

(Answer five full questions, each carries two marks)

- 1 Define Normality of a Solution
- 2 List the various systems of units
- 3 Define Conduction
- 4 What is distillation
- 5 Define Azeotrope

(5X2 = 10)

II

PART-B

(Answer any five questions, each carries six marks)

- 1 Describe the various systems of units and the fundamental quantities associated with them.
- 2 Calculate the amount of NaOH required to prepare one litre of
a) 2N solution b) 3 Molar solution
- 3 Derive the equation for heat transfer by conduction through Plane wall.
- 4 Distinguish between Steady state and Unsteady state conduction.
- 5 Describe the unit operations a) Absorption b) Evaporation
- 6 Differentiate between Distillation and Extraction
- 7 Explain a chemical process with flow diagram.

(5X6 = 30)

PART - C

(Write one full question from each unit)

(15X4=60)

UNIT-I

- III a)** Explain the concept of Unit Operation and Unit Process with two examples for each. **8**
- b) Calculate the Normality of Sodium Carbonate and HCl using the following data:
a) 1.3 gm of Sodium Carbonate is dissolved in 250ml Water b) 20ml of Sodium Carbonate solution is neutralised by 23ml of Hydrochloric acid solution. **7**

OR

- IV a)** Calculate the volume corresponds to one gram of Nitrogen at 2 atm pressure and 700 Degree Kelvin **8**
- b) Using Dimensional Analysis convert Joule into Erg. **7**

UNIT-II

- V a)** Derive the expression for heat transfer through furnace wall made of three different materials in series. **8**
- b) Calculate the heat loss per unit area of a brick wall of thickness 0.6m, Inner surface Temp = 410 degree kelvin, Outside surface temp = 320 degree kelvin. The thermal conductivity of brick may be taken as 0.7W/(m.K) **7**

OR

- VI a)** Derive an expression for heat flow through thick walled cylinder by conduction. **8**
- b) Explain the temp gradient in forced convection. **7**

UNIT-III

- VII a)** Explain with fig, the working of Vertical Ballman Extractor to extract Cotton seed oil **8**
- b) Explain horizontal and vertical tube Evaporator **7**

OR

- VIII a)** Explain with fig Double pipe heat Exchanger **8**
- b) Compare Absorption and Distillation **7**

UNIT-IV

- IX a)** Explain with fig, the working of a Packed Column for Distillation **8**
- b) Explain the operation of a Distillation Column under Total Reflux **7**

OR

- X a)** Explain with figure Simple Distillation **8**
- b) Explain the operation of a Distillation Column under minimum Reflux **7**